Progress Log — Pandora Paradox

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This document records the work I have done on top of the project as described in README.md. All original files have been kept intact; everything below is additive.

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1. Starting point

As described in README.md, the project targets the question

> "Can a film's performance metrics predict its cultural impact?"

The repository defined a 6-phase pipeline:

Phase	Script	Role
1	`01_collect_movies.py`	Scrape top-200 films from Wikipedia
2	`02_cultural_footprint.py`	Collect Wikipedia / Reddit / Trends metrics
3	`03_movie_scripts.py`	Scrape and analyse film scripts
4	`04_build_index.py`	Integrate everything into a Cultural Footprint Index
5	`05_ml_model.py`	Train regression + classification models on the CFI
6	`06_live_data_collectors.py`	Real API collectors for Wikipedia / Reddit / TMDB / Trends / IMSDb

The included `data/model_results.json` reported **R^2 = 0.10** / **AUC = 0.55** — both close to random.

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2. Initial audit of the existing codebase

Before touching anything I read through the scripts and found that the pipeline was running on randomly-generated sample data rather than real data:

- **`02_cultural_footprint.py`**: a hand-authored dictionary holds values for ~30 famous films; for the remaining 170 titles, genre-based `np.random.normal()` calls fabricate cultural metrics.
- **`03_movie_scripts.py`**: the same pattern — hardcoded NLP feature values for a handful of films, then genre-conditioned random draws for everything else. No actual script text is parsed.
- **`06_live_data_collectors.py`**: fully implemented real-API code (Wikipedia pageviews + language editions, Reddit search, TMDB, Google Trends, IMSDb scraping) — but never actually run by the existing pipeline.
- **Consequence**: the R^2 / AUC numbers in `model_results.json` reflect that sample data, not real data.

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3. Bug fixes to `06_live_data_collectors.py`

Two latent bugs prevented the existing live collectors from working correctly:

3.1 Wikipedia pageviews — title candidate fallback

- The URL builder always appended the disambiguation suffix `_(YYYY_film)`. This works for *Avatar_(2009_film)* but fails for *Avengers:_Endgame* (which does not use disambiguation).
- Fix: now tries multiple candidate titles in order — `Title_(YYYY_film)` → `Title` → `Title_(film)` — and uses the first one that returns 200 OK.
- Also properly URL-encodes spaces to underscores.

3.2 IMSDb scraper — URL pattern matching

- Was constructing `/scripts/title-YYYY.html` (lowercase + year). Actual IMSDb URLs do not include the year and often use PascalCase or the "Title, The" convention.
- Fix: tries several variants (`Title-Name`, `Title-Name,-The`, `TitleName`, `title-name`) and picks the first 200 OK that isn't a "not found" page.

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4. New scripts added

The original files were left unchanged. Four new scripts and one helper sit alongside them:

New file	What it does
`scripts/03_1_subtitle_features.py`	Scrapes YIFY subtitles by IMDb ID (via TMDB `external_ids`), parses SRTs, computes dialogue-level NLP features
`scripts/04_1_build_index.py`	Flattens `live_api_data.json` + `extended_api_data.json` + subtitle features into a real-data Cultural Footprint Index
`scripts/05_1_ml_model.py`	Trains Model A ("strict pre-release") and Model B ("+ early reception"); reports R^2, AUC, and feature importance
`scripts/06_1_extended_collectors.py`	Adds OMDb awards, Wikiquote realised quotability, and Reddit-derived meme signals
`scripts/export_summary_xlsx.py`	Generates `pandora_database_summary.xlsx` (two-tab bilingual summary)

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5. Data sources integrated

Source	Purpose	API key needed	Coverage
Wikipedia pageviews (Wikimedia REST)	Long-term reference interest	no	98 %
Wikipedia language editions	Global reach	no	100 %
Reddit public search	Discussion volume / engagement	no	99 %
TMDB	Budget / revenue / popularity / rating	yes (free)	98 %

Google Trends (pytrends)	Search interest	no	**7 %** — rate-limited
IMSDb	Script structure (supplementary)	no	22 %
YIFY Subtitles (new)	Dialogue NLP — replaces IMSDb as primary script source	no	**95 %**
OMDb API (new)	Awards, IMDb rating / votes, Metascore	yes (free)	96 %
Wikiquote (new)	Realised quotability — replaces blocked IMDb quotes	no	93 %
Reddit-derived (new)	Meme penetration, subreddit diversity	computed	100 %

IMDb direct scraping was attempted but returns HTTP 202 + empty body (anti-bot). Wikiquote + OMDb together cover the same research need.

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6. New features designed

6.1 Meme-prediction predictors (X-side, subtitle-derived)

- **`short_punchy_count`** — lines of 2-6 words ending in `.` / `!` with no `?` (structural quotability potential)
- **`short_punchy_density`** — the above, normalised by total dialogue lines
- **`rare_proper_noun_count`** — mid-sentence capitalised tokens appearing ≤ 2 times (invented names like Pandora, Thanos)
- **`sentiment_spike`** — largest absolute sentiment change between consecutive lines
- **`sentiment_peak`** — maximum absolute line-level sentiment

6.2 Cultural-impact outcomes (Y-side)

- **`wikiquote_count`** — realised quotability from Wikiquote
- **`meme_post_count`** — posts in meme-specific subreddits (derived from Reddit data already collected)
- **`subreddit_diversity`** — number of distinct subreddits discussing the film
- **`subreddit_concentration`** — Herfindahl index of subreddit distribution

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7. Pipeline executed on all 197 films

Both collectors ran in parallel in the background. Final coverage:

Collector	Success rate
Wikipedia pageviews	193 / 197 (98 %)
Wikipedia language editions	197 / 197 (100 %)
Reddit mentions	196 / 197 (99 %)
TMDB metadata	194 / 197 (98 %)
YIFY subtitles	187 / 197 (95 %)

IMSDb scripts	43 / 197 (22 %)
Google Trends	13 / 197 (7 %)
OMDb awards	190 / 197 (96 %)
Wikiquote	183 / 197 (93 %)
Reddit-derived	197 / 197 (100 %)

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8. CFI rebuilt on real data

The original CFI weighted seven components, of which four (`quotability_score`, `meme_score`, `merchandise_index`, `awards`) were hand-coded sample values. The new CFI uses eight **all-observable** components:

Component	Weight	Real source
Wikipedia views	20 %	Wikimedia API
Wikipedia languages	10 %	Wikimedia API
Reddit engagement	20 %	Reddit API
TMDB vote count	10 %	TMDB API
Google Trends	10 %	pytrends (7 % populated)
Wikiquote count	10 %	Wikiquote scrape
Awards (wins + noms)	10 %	OMDb
Meme-subreddit posts	10 %	Derived from Reddit

`cfi_score` (0-100) and `cfi_residual` (residual from `log(gross) + log(budget)` regression) are stored per film.

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9. First real-data ML results

Model A — strict pre-release (metadata + subtitle NLP only)

- Regression: test R^2 = +0.026, CV R^2 = –0.236 ± 0.13
- Classification (overperformer vs underperformer): best CV AUC 0.636 (Random Forest)

Model B — + early reception (tmdb_popularity, imdb_rating, metascore)

- Regression: test R^2 = +0.102, CV R^2 = +0.070 ± 0.18
- Classification: best CV AUC 0.725 (Random Forest)

Top Model A features (regression)

1. `rare_proper_noun_count` 0.120 ← newly designed
2. `short_punchy_count` 0.093 ← newly designed

3. `humor_indicator` 0.067
4. `vocabulary_richness` 0.064
5. `short_punchy_density` 0.057

The two peak-based features I designed top the list, ahead of budget, sequel flag, or franchise flag. This directly validates the hypothesis that extractable units matter more than averages for meme prediction.

9.1 `years_since_release` experiment — null result

After the first run I added `years_since_release` (current year – release year) as a metadata feature, on the hypothesis that older films should accumulate more Wiki views / Reddit posts / etc.

Re-running 04_1 → 05_1 with the feature included:

Metric	Before	After	Change
Model A regression test R^2	+0.026	−0.016	slight ↓
Model A regression CV R^2	−0.236	−0.308	↓
Model A best classifier CV AUC	0.636	0.657	+0.02
Model B regression test R^2	+0.102	+0.208	↑
Model B regression CV R^2	+0.070	−0.042	↓
Model B best classifier CV AUC	0.725	0.729	flat

`years_since_release` did **not** appear in the top-10 feature importances of either model. The likely reason is that the new CFI is structurally time-controlled: Wikipedia pageviews use a rolling 3-year window, Reddit returns "current" hot posts, Wikiquote counts are editor-curated rather than time-accumulated, and award counts are mostly fixed at release. So there was little time-bias left to absorb. The feature was retained in the codebase as a recorded experiment.

CV variance was high ($\sigma \approx 0.21$–0.30) — the test-R^2 swings between runs are largely noise from a small ($n = 197$) sample, not real signal.

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10. Output files

Data

- `data/live_api_data.json` — raw live-API results for 197 films
- `data/extended_api_data.json` — live data + OMDb + Wikiquote + Reddit-derived
- `data/subtitle_features.csv` — NLP features from real subtitles
- `data/cultural_footprint_real.csv` — CFI components
- `data/pandora_full_dataset_real.csv` — full merged dataset (87 columns)
- `data/model_results_real.json` — Model A / B metrics and feature importance

Deliverables / documents

- `pandora_database_summary.xlsx` — two-tab summary (JA / EN)
- `meeting_brief_2026-04-22.md` / `.pdf` — English meeting brief
- `meeting_brief_2026-04-22.ja.md` / `.ja.pdf` — Japanese meeting brief

- `PROGRESS.md` / `.ja.md` — this log

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11. What is scheduled next

For the 2026-04-26 intermediate deadline (dataset only):

- 4/23 — add `years_since_release`, fix OMDb year-mismatch edge cases, write and run YouTube trailer collector
- 4/24 — run visual-distinctiveness feature extractor on posters, final CFI re-run
- 4/25 — data integration + presentation-video recording
- 4/26 — buffer day and intermediate submission

Post-deadline:

- Analysis notebook refresh on real data
- Paradox case studies (top residuals)
- Feature-importance visualisation
- CFI-weight sensitivity analysis
- Final report and slides